

# BattleBots App

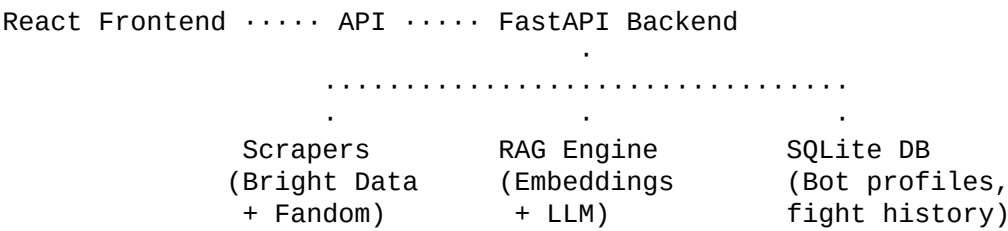
An **AI-powered BattleBots fight predictor** with web scraping, RAG (Retrieval-Augmented Generation), and LLM-based fight analysis · full-stack architecture with FastAPI backend and React frontend.

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## · Features

- **Bot Scraping** · Collect bot profiles from BattleBots data + Fandom wiki
  - **RAG Pipeline** · Embed bot data for context-rich predictions
  - **Fight Predictions** · LLM-powered outcome predictions with reasoning
  - **Full-Stack** · FastAPI backend + React/TypeScript frontend
  - **SQLite** · Local database for bot profiles and fight history
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## .. Architecture



## · File Structure

```
BattleBotsApp/
... backend/
  ... main.py                # FastAPI app with CORS, routers, lifespan
  ... scrapers/scrape_bots.py # Bright Data + Fandom wiki scraper
  ... rag/                   # RAG embedding and retrieval
  ... routes/                # API route handlers
... frontend/
  ... src/                   # React TypeScript app
  ... package.json
  ... pyproject.toml
```

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## · Getting Started

```
# Backend
cd backend && uv sync && uv run python main.py
# Frontend
cd frontend && npm install && npm run dev
```

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## · Logic Flow

- 1 **Scrape** · Bot profiles collected from web sources
  - 2 **Store** · Bot data persisted to SQLite
  - 3 **Embed** · Profiles embedded into vector store
  - 4 **Select** · User picks two bots for fight prediction
  - 5 **Retrieve** · Relevant context from vector store
  - 6 **Predict** · LLM analyzes bots and predicts outcome
  - 7 **Display** · Prediction shown in frontend
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## · Dependencies

**Backend:** FastAPI, OpenAI, Anthropic, BeautifulSoup4, SQLAlchemy **Frontend:** React, TypeScript, Vite

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## · License

Educational project · use freely for learning and reference.